STRATEGIC AI GOVERNANCE FOR ENTERPRISE CYBERSECURITY

APPENDIX I

Artifact Evaluation — Benchmark, Significance, Stability and Explainability

Evidence that the integrated Graph Intelligence + Safe RL + XAI artifact described in Part III was built and evaluated

Output Images:

Results:

```
--- Objective 2/4: Evaluation on the ORIGINAL (imbalanced) distribution ---
    (We train on balanced data but evaluate on the real distribution — the honest test.)

  DQN  | Accuracy: 0.9760 | Precision: 0.9657 | Recall: 0.9121 | F1: 0.9382
  PPO  | Accuracy: 0.9391 | Precision: 0.8010 | Recall: 0.9251 | F1: 0.8586
```

Agent Performance (Evaluated on Full 500,000-Flow Subsample)

```
[1/2] Evaluating DQN agent on full dataset...

--- DQN ---
     Accuracy : 0.9760
     Precision: 0.9657
     Recall   : 0.9121
     F1-score : 0.9382

     Per-class report:
                  precision    recall  f1-score   support

          Normal       0.98      0.99      0.99    400104
          Attack       0.97      0.91      0.94     99896

        accuracy                           0.98    500000
       macro avg       0.97      0.95      0.96    500000
    weighted avg       0.98      0.98      0.98    500000

[2/2] Evaluating PPO agent on full dataset...
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203:
  return datetime.utcnow().replace(tzinfo=utc)

--- PPO ---
     Accuracy : 0.9391
     Precision: 0.8010
     Recall   : 0.9251
     F1-score : 0.8586

     Per-class report:
                  precision    recall  f1-score   support

          Normal       0.98      0.94      0.96    400104
          Attack       0.80      0.93      0.86     99896

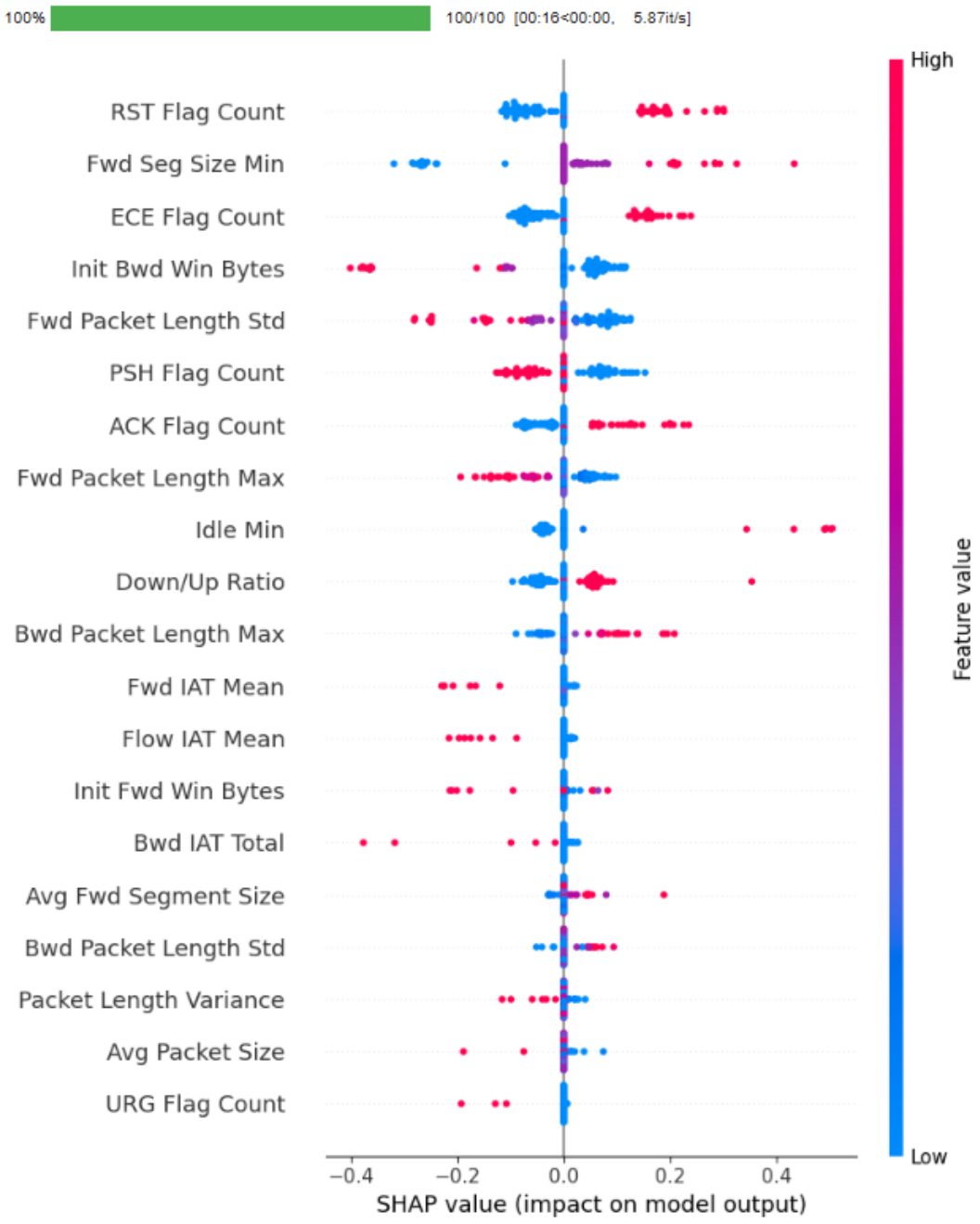
        accuracy                           0.94    500000
       macro avg       0.89      0.93      0.91    500000
    weighted avg       0.94      0.94      0.94    500000

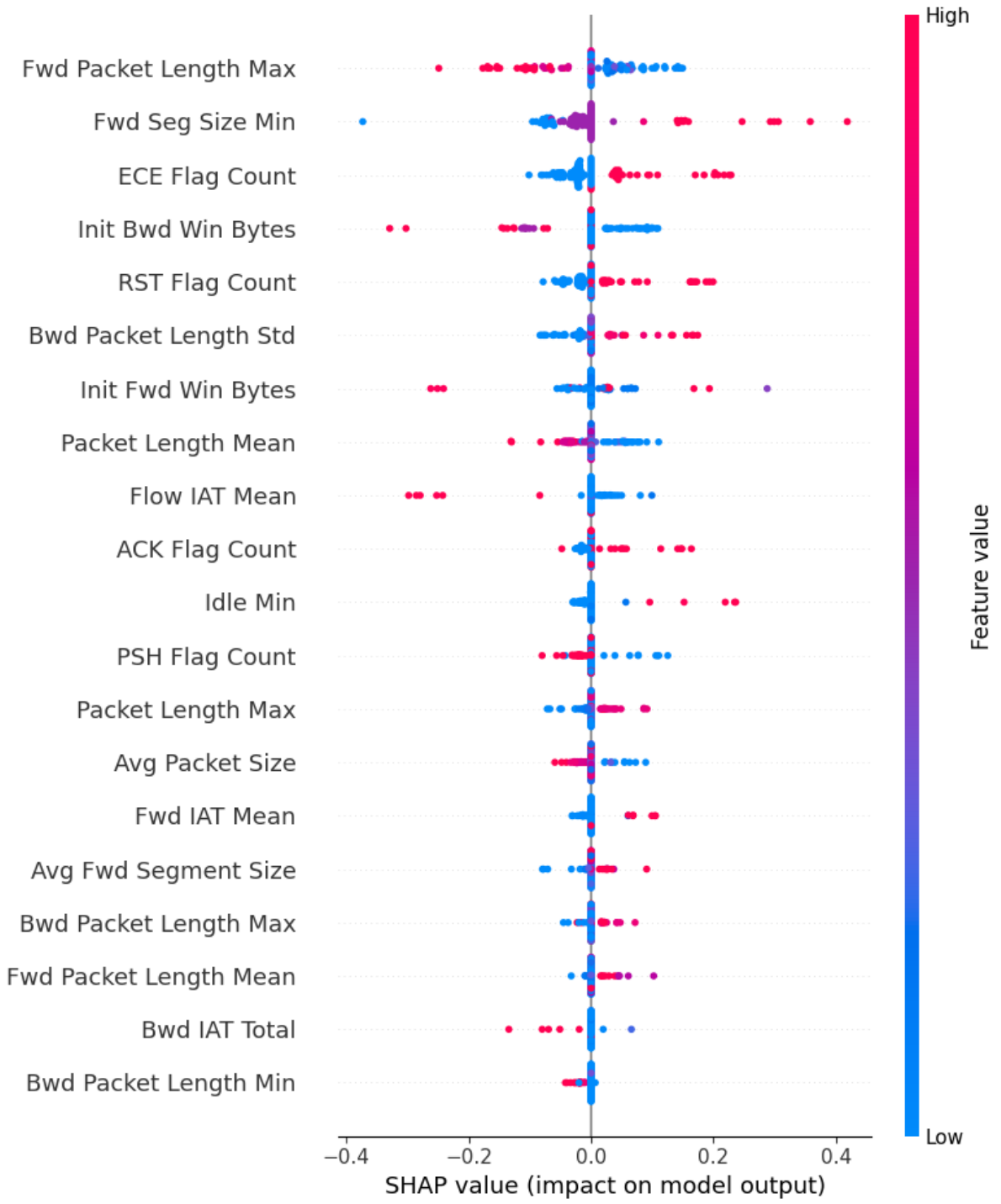
--- Summary ---
     Metric            DQN        PPO
     accuracy       0.9760     0.9391
     precision      0.9657     0.8010
     recall         0.9121     0.9251
     f1             0.9382     0.8586
```

```
🔄 Building class-balanced TRAINING set...
   Before balancing -> normal: 400,104 | attack: 99,896
   SMOTE applied.
   After balancing  -> normal: 400,104 | attack: 400,104
```

SHAP Global Feature Importance: Top 20 Features by Rank

Explainability Integration Output from Python code:




Fwd Packet Length Max
Fwd Seg Size Min
ECE Flag Count
Init Bwd Win Bytes
RST Flag Count
Bwd Packet Length Std
Init Fwd Win Bytes
Packet Length Mean
Flow IAT Mean
ACK Flag Count
Idle Min
PSH Flag Count
Packet Length Max
Avg Packet Size
Fwd IAT Mean
Avg Fwd Segment Size
Bwd Packet Length Max
Fwd Packet Length Mean
Bwd IAT Total
Bwd Packet Length Min
−0.4
−0.2
0.0
0.2
0.4
SHAP value (impact on model output)
High
Feature value
Low